

IBRAHIM ARSHAD

AI Systems Engineer

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SUMMARY

AI systems engineer with 5+ years of experience turning product requirements into production systems, with a focus on reliable agents, evals, infrastructure, distributed systems, and blockchain protocols.

SELECTED WORK

SettleDiff

Transaction forensics for agent purchases

- Built an evidence-first system for reconstructing what an agent purchase promised, executed, and actually settled.
- Added deterministic checks across payment execution and settlement evidence instead of relying only on provider receipts.
- Integrated Context.dev and Hyperfusion for external evidence and explanation workflows.

Python · **Perflo** · **x402** · **EIP-3009** · **Context.dev** · **Hyperfusion**

CaseZero

Evidence-first AI investigations

- Building an AI investigation system that reasons over public evidence while withholding the official conclusion until the generated assessment is locked.
- Records evidence sets, model versions, prompt versions, and assessment hashes so hindsight and reasoning changes can be measured.
- Uses structured evidence workflows rather than unconstrained agent output.

AI agents · **evals** · **structured evidence** · **Context.dev** · **Hyperfusion**

Vault Steward

Local-first integrity system for Obsidian

- Built and published an Obsidian plugin that combines deterministic checks with bounded model-assisted review.
- Detects broken references, schema issues, contradictions, stale knowledge, orphaned tasks, and unresolved decisions.
- Keeps vault data local and requires explicit user approval before applying proposed edits.

TypeScript · **Obsidian** · **Ollama** · **OpenAI** · **agent workflows** · **evals**

EXPERIENCE

Hypotenuse Labs

Remote

Software Engineer

Jan 2025 – Present

- Lead client projects end-to-end, translating stakeholder requirements into technical scope, system architecture, and production delivery.
- Design product and system architecture across backend, AI, and protocol-heavy applications.
- Built a deterministic Rust perpetual futures engine covering order matching, margin, funding, oracle safety, and liquidations.
- Built a Rust zk-rollup prototype for policy-constrained, provable AI workflow execution.
- Built an MCP-based integration gateway for AI providers and tools.
- Built Stellar vault and reward infrastructure plus Merkle-based reward services.
- Deploy and operate production backend infrastructure on AWS.

Lucia Protocol

Remote

Blockchain Engineer

Apr 2024 – Dec 2024

- Built backend services and APIs for distributed financial products and transaction processing.
- Collaborated on product architecture and system design from requirements through implementation.
- Integrated backend services with blockchain networks and external systems.

Fantom Foundation

Remote

Blockchain Developer

Aug 2021 – Apr 2024

- Built backend infrastructure, indexing pipelines, GraphQL services, and APIs for production blockchain applications.
- Designed systems for high-volume event processing and data consistency.
- Improved backend reliability through fault-tolerant architecture and observability.
- Performed smart-contract security reviews and supported ecosystem production deployments.

OPEN SOURCE CONTRIBUTIONS

- **vLLM** · merged prefill-path reliability fix
- **LangGraphJS** · merged RemoteGraph checkpoint-state fix
- Additional contributions across **CrewAI**, **Reth**, **Alloy**, **Qdrant**, and **Aida**, covering agent execution, inference, transaction lifecycle, storage, and infrastructure reliability.

TECHNICAL

Languages: Rust, Python, TypeScript, Solidity

AI Systems: Agents, evals, MCP, agent harnesses, structured outputs, RAG

Product & Systems: Technical scoping, product architecture, system design, API design

Infrastructure: AWS, Docker, GitHub Actions, CI/CD

EDUCATION

BSc Computer Science

2015–2019

University of Wollongong Dubai